



Chapter 2

Digital Image Fundamentals

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Digital Image Fundamentals

- 2.1 Elements of Visual Perception
- 2.2 Light & EM Spectrum
- 2.3 Image Sensing & Acquisition

2.3.4 A simple Image Formation Model

- $f(x,y) = i(x,y) r(x,y)$
 - $i(x,y)$: illumination components, $0 < i(x,y) < \infty$
 - $r(x,y)$: reflectance components, $0 < r(x,y) < 1$

2.4.1 Basic Concept in Sampling & Quantization

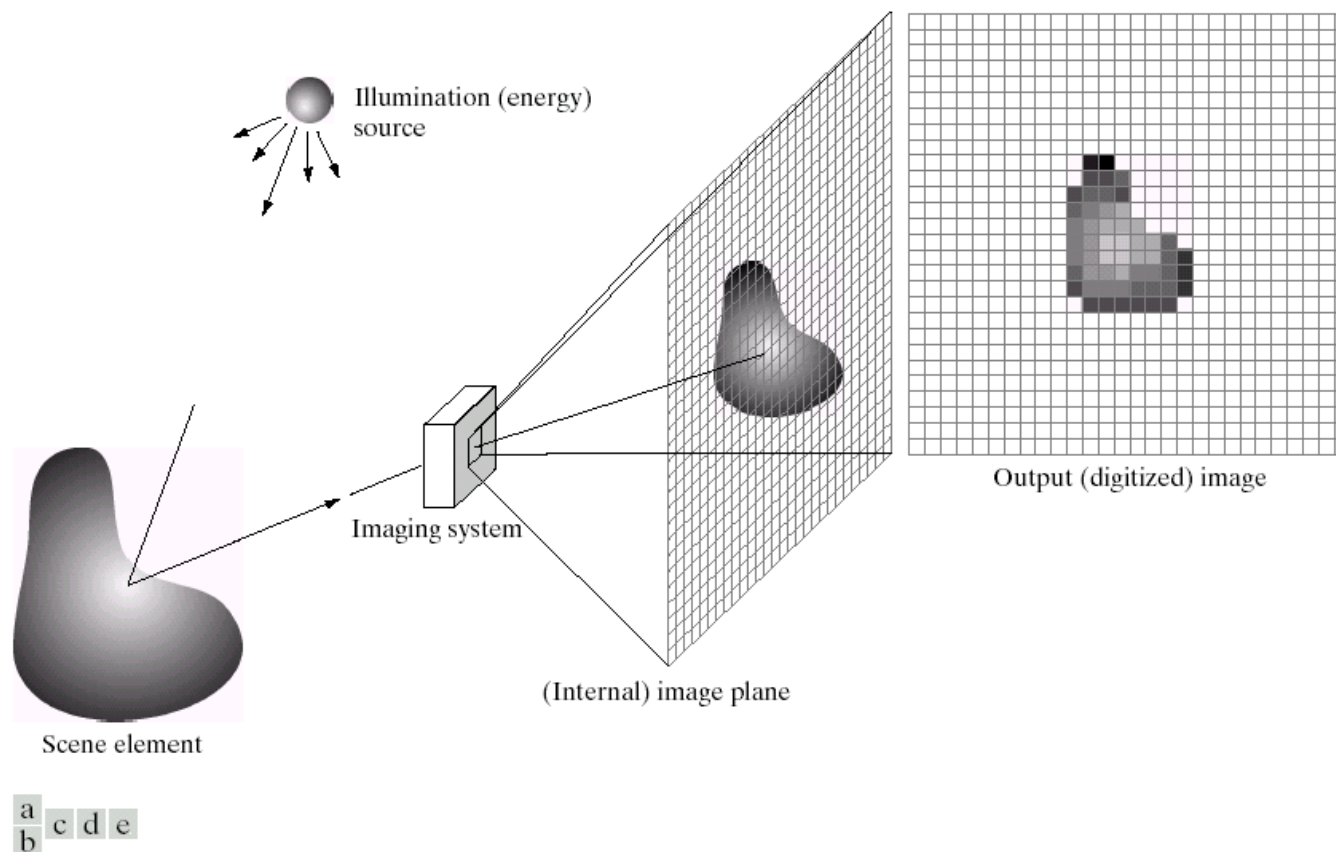
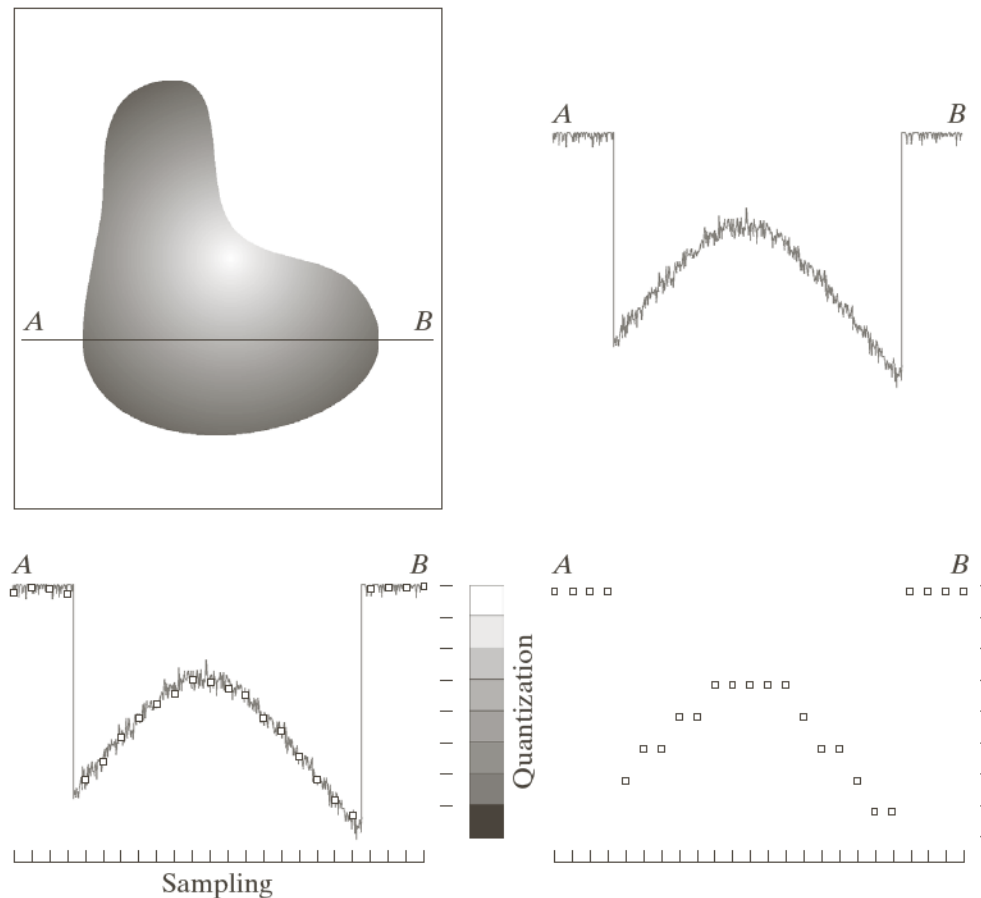


FIGURE 2.15 An example of the digital image acquisition process. (a) Energy ("illumination") source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

2.4.1 Basic Concept in Sampling & Quantization



a	b
c	d

FIGURE 2.16

Generating a digital image. (a) Continuous image. (b) A scan line from A to B in the continuous image, used to illustrate the concepts of sampling and quantization. (c) Sampling and quantization. (d) Digital scan line.

2.4.1 Basic Concept in Sampling & Quantization

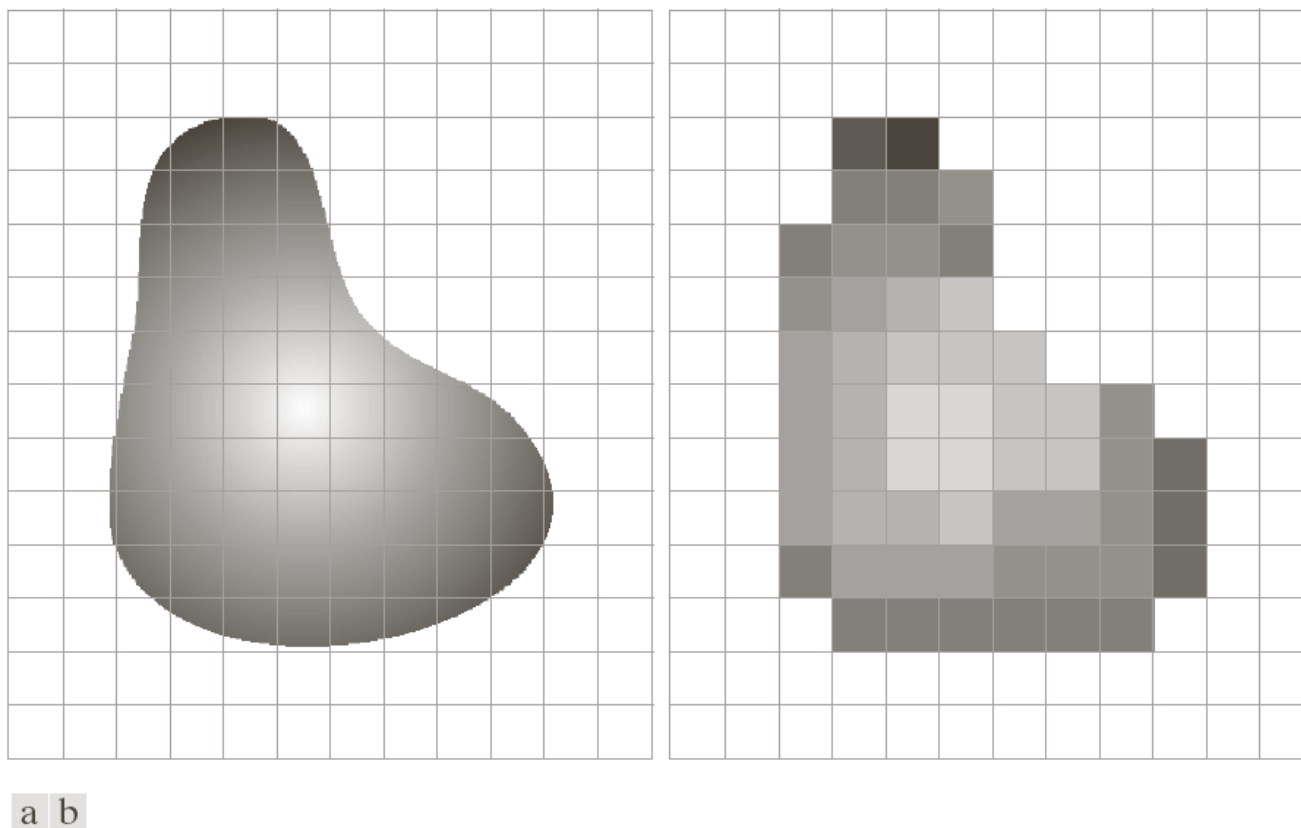


FIGURE 2.17 (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.

2.4.2 Representing Digital Image

- Matrix or 2-D Array

2.4.3 Spatial & Intensity Resolution

- Spatial Resolution
 - Line pairs per unit distance
 - Dots (pixels) per unit distance
 - Dpi: dot per inch
 - **Spatial resolution** is a measure of the accuracy or detail of a graphic display, expressed as dots per inch, pixels per inch, lines per inch, etc.

Spatial Resolution



a b
c d

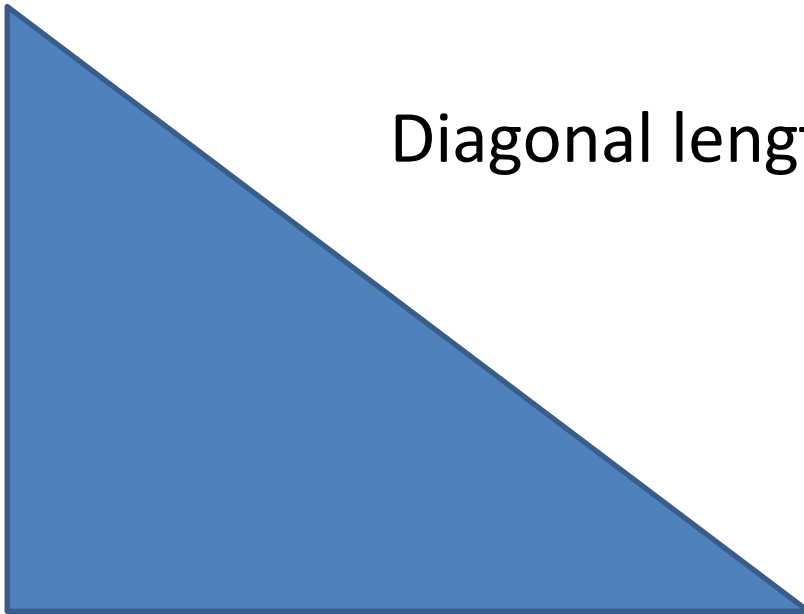
FIGURE 2.20 Typical effects of reducing spatial resolution. Images shown at: (a) 1250 dpi, (b) 300 dpi, (c) 150 dpi, and (d) 72 dpi. The thin black borders were added for clarity. They are not part of the data.

Exercise

- 10 inch , 1024 X 768
- Spatial Resolution =?



Exercise



Diagonal length = 10 inch

2.4.3 Spatial & Intensity Resolution

- Intensity Resolution
 - Intensity Level
 - Usually is an integer power of two
 - 8 bits
 - HDMI 1.3 support 10bits,12bits,16bits for each color component (channel)

Intensity Resolution

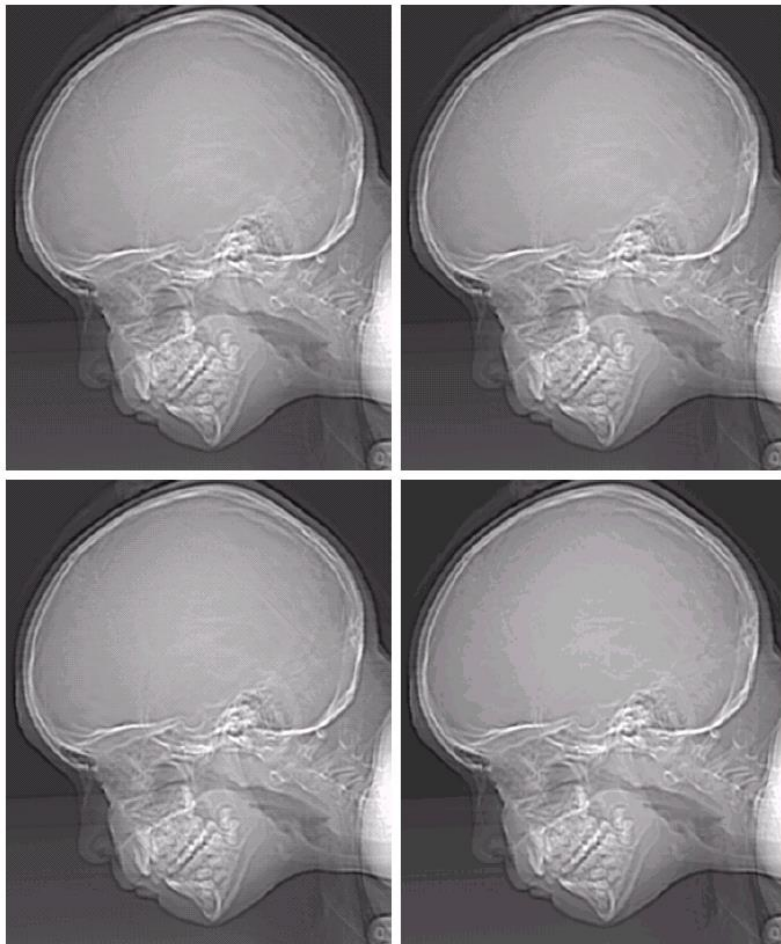
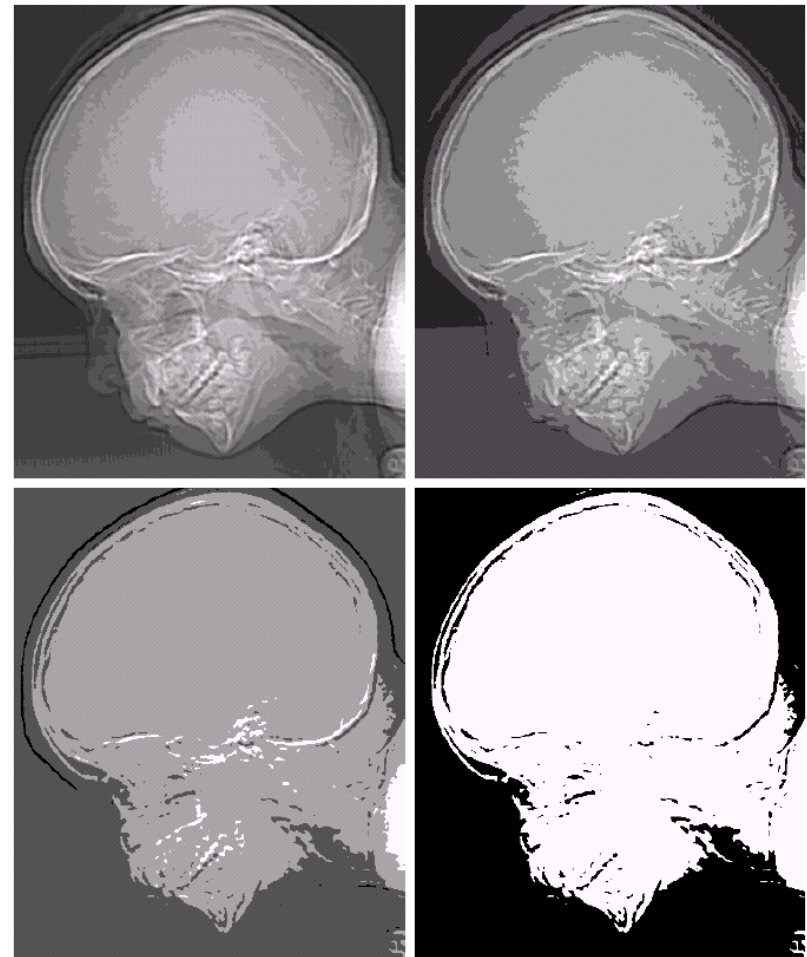


FIGURE 2.21
(Continued)
(e)–(h) Image displayed in 16, 8, 4, and 2 gray levels. (Original courtesy of Dr. David R. Pickens, Department of Radiology & Radiological Sciences, Vanderbilt University Medical Center.)



Intensity Resolution

- How to get n-bits image?



Intensity Resolution

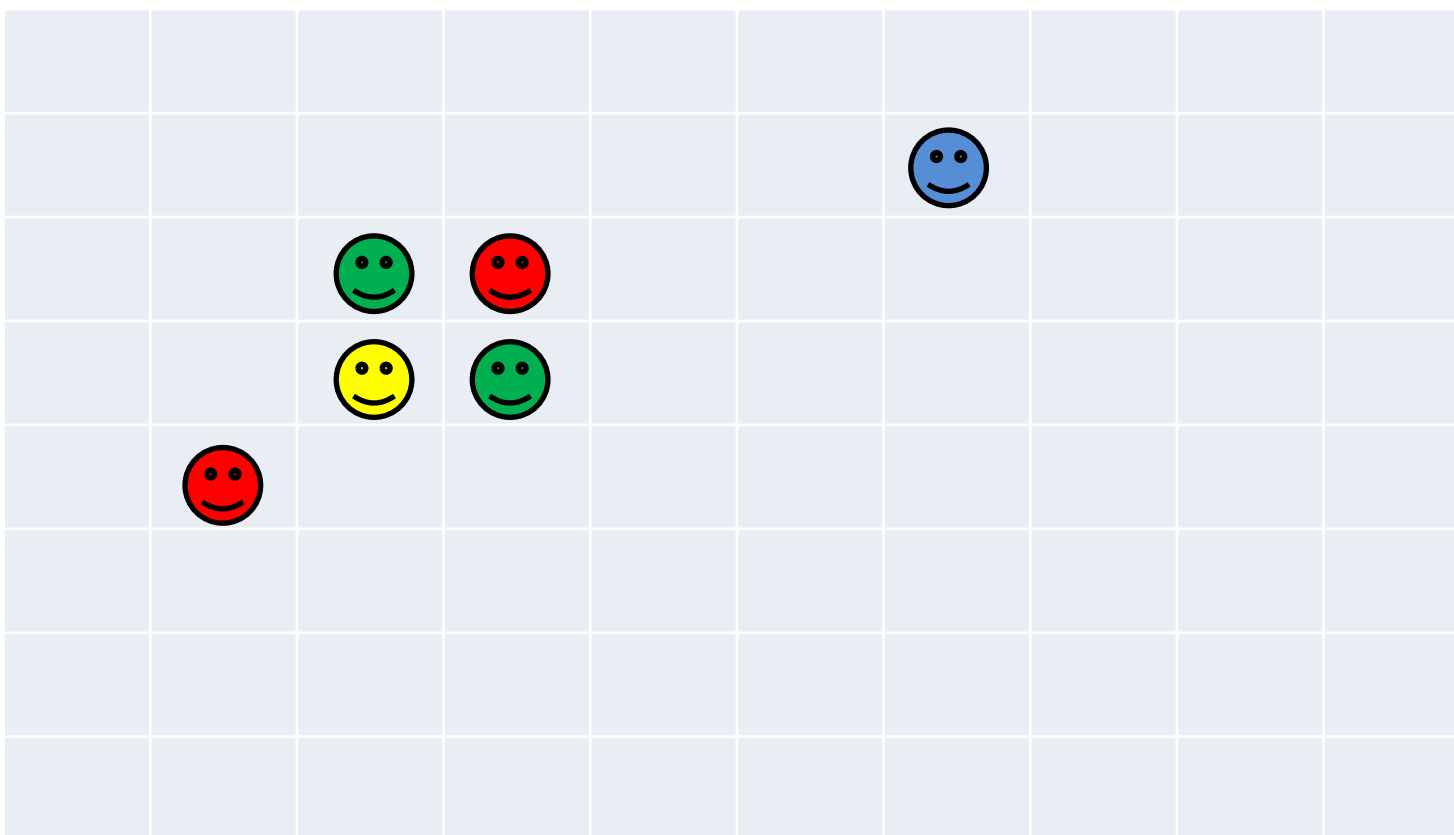
- [Programming]

- **Exercise:**

```
byte[,] get_n_bits_image(byte[,] f,int n);
```

2.5 Same Basic Relationship between pixels

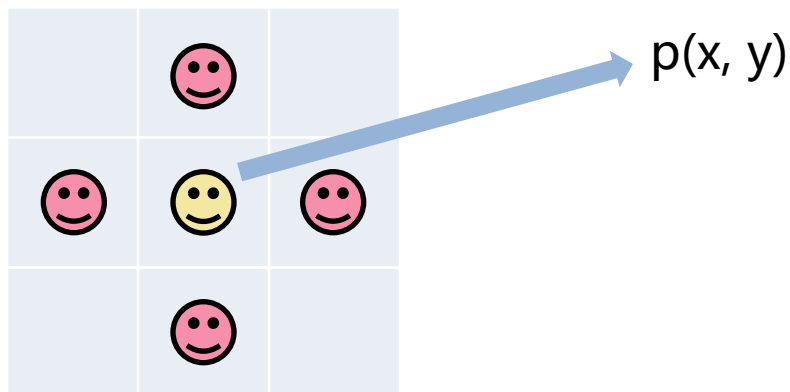
2.5.1 Neighbors of a pixel



2.5.1 Neighbors of a pixel

- 4-Neighbors of $p(x, y)$:

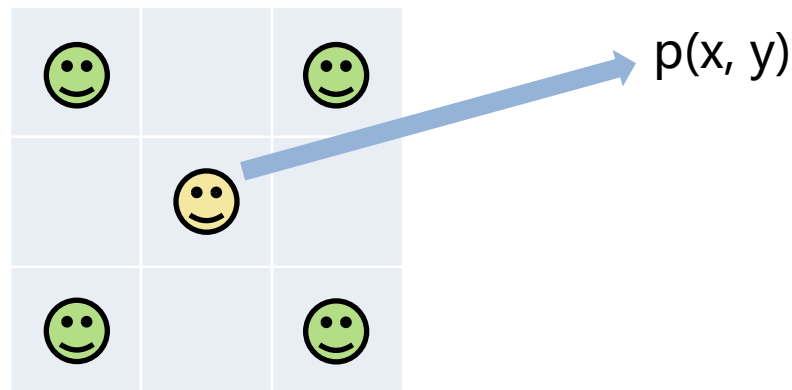
$$N_4(p) = \{(x+1, y), (x-1, y), (x, y+1), (x, y-1)\}$$



2.5.1 Neighbors of a pixel

- D-Neighbors (Diagonal -Neighbors) of $p(x, y)$:

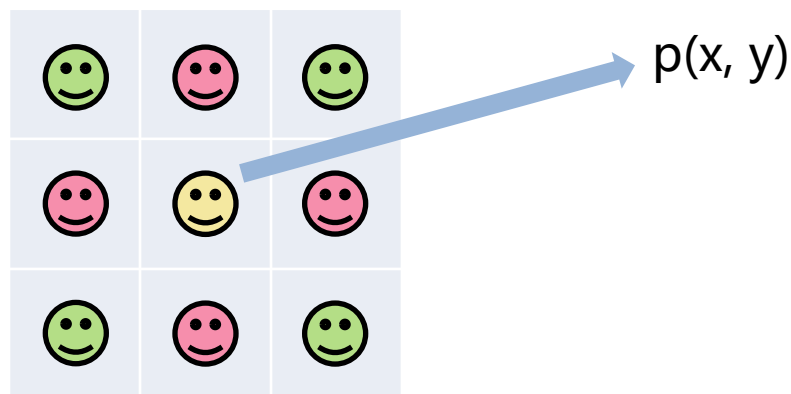
$$N_D(p) = \{(x+1, y+1), (x-1, y+1), (x+1, y-1), (x-1, y-1)\}$$



2.5.1 Neighbors of a pixel

- 8-Neighbors of $p(x, y)$:

$$N_8(p) = N_4(p) \cup N_D(p)$$



2.5.2 Adjacency, Connectivity, Region and Boundaries

- Adjacency

Binary image: Intensity $\{0,1\}$

Usually 0 means background and 1 means foreground

0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	1	1	0	0	0
0	0	1	1	1	1	0	0
0	0	1	1	1	1	0	0
0	0	1	1	1	1	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

Adjacency

- Both p and q are in the foreground ($V:\{1\}$)
 - (a) 4-adjacency: q in $N_4(p)$
 - (b) 8-adjacency: q in $N_8(p)$
 - (c) m-adjacency (mixed adjacency)
 - (i) q in $N_4(p)$ or
 - (ii) q in $N_8(p)$ and the set $N_4(p) \cap N_4(q)$ has no pixels whose value are from $V(V:\{1\})$

Adjacency

(a) m-adjacency (mixed adjacency)

- (i) q in $N_4(p)$ or
- (ii) q in $N_8(p)$ and the set $N_4(p) \cap N_4(q)$ has no pixels whose value are from $V \setminus \{1\}$

0		
0		0
0	0	

0	1	1
0	1	0
0	0	1

m-Adjacency

- p: (yellow), q: (green)
- Blue background pixel means
 - set $N4(p) \cap N4(q)$
- (a) m-adjacency (mixed adjacency)
 - q in $N4(p)$ or
 - q in $N8(p)$ and the set $N4(p) \cap N4(q)$ has no pixels whose value are from $V(V:\{1\})$

0	0	0	0
0			0
0		0	0
0	0		0

0	0	0	0
0			0
0		0	0
0	0		0

0		
0		0
0	0	

m-Adjacency

- p : (yellow), q : (blue)
- Blue background pixel means
 - set $N4(p) \cap N4(q)$
- (a) m-adjacency (mixed adjacency)
 - (i) q in $N4(p)$ or
 - (ii) q in $N8(p)$ and the set $N4(p) \cap N4(q)$ has no pixels whose value are from $V \setminus \{1\}$

0		
0		0
0	0	

[Programing]

- `bool Is_4adjacent(int px,int py,int qx,int qy);`
- `bool Is_8adjacent(int px,int py,int qx,int qy);`
- `Bool Is_m_adjacent(byte[,]f, int px,int py,int qx,int qy);`

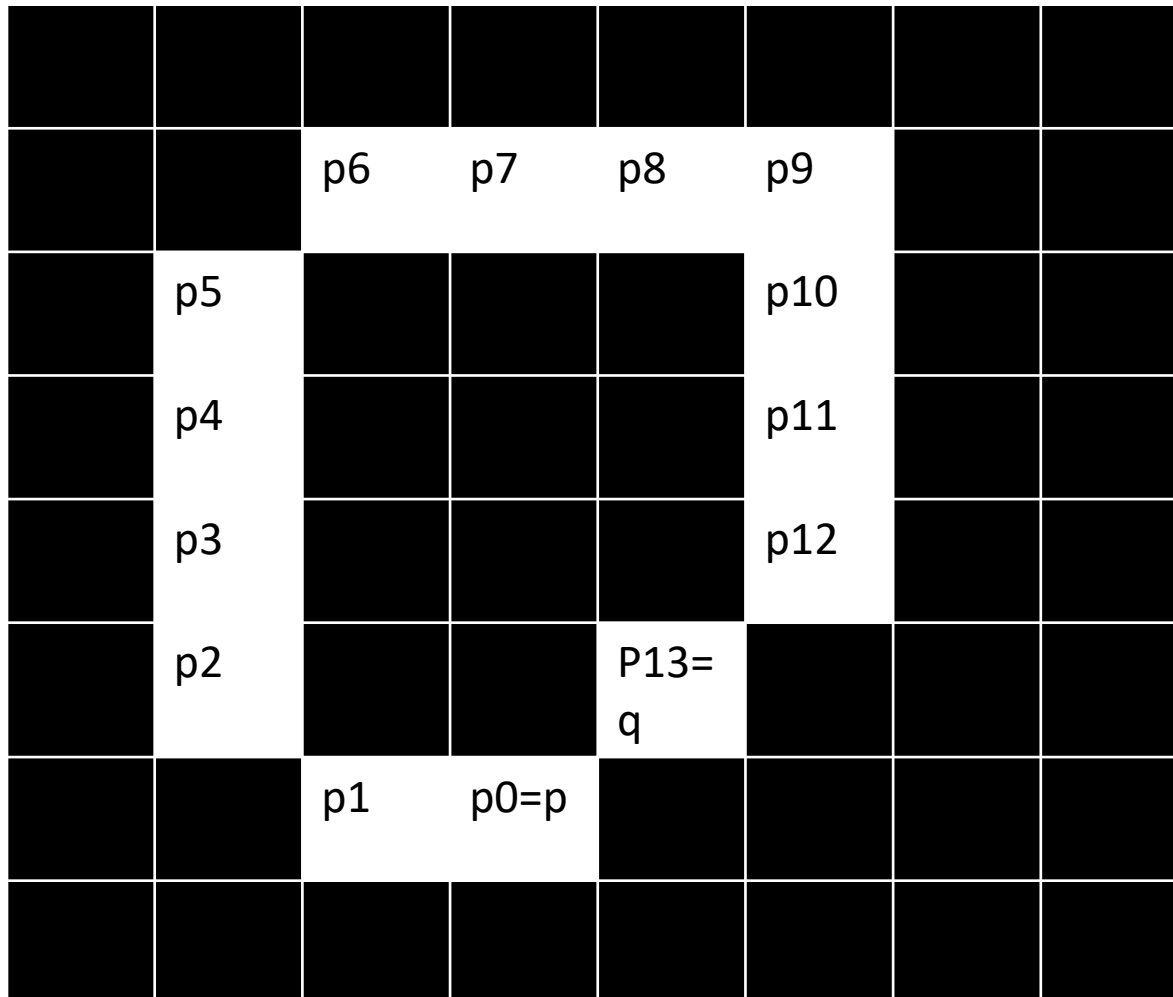
Digital Path

- Digital Path from pixel p to pixel q is a sequence of distinct pixels with coordinates $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$ where (x_{i-1}, y_{i-1}) and (x_i, y_i) are adjacent, $i: [1..n]$

Digital Path

		p6	p7	p8	p9=q		
	p5						
	p4						
	p3						
	p2						
		p1	p0=p				

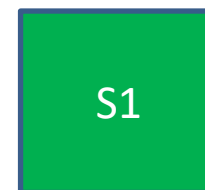
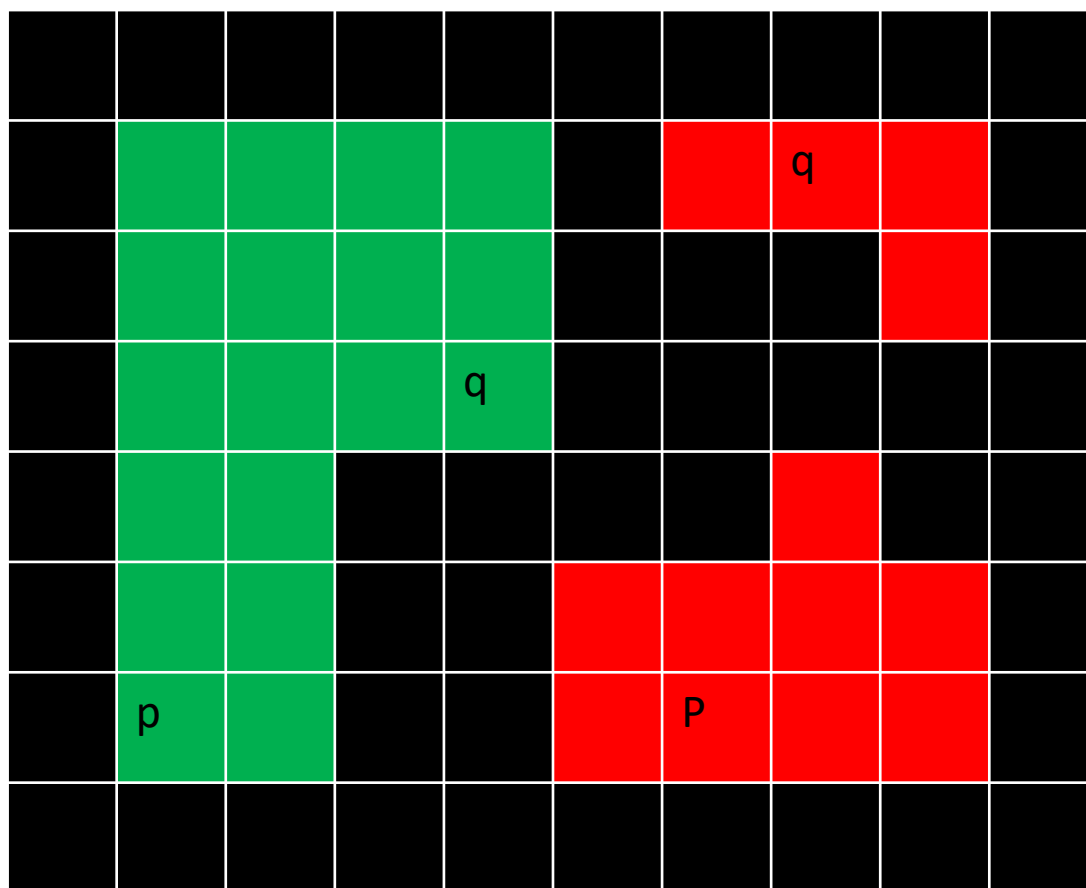
Digital Path (Closed)



Connected Set (Region)

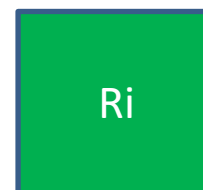
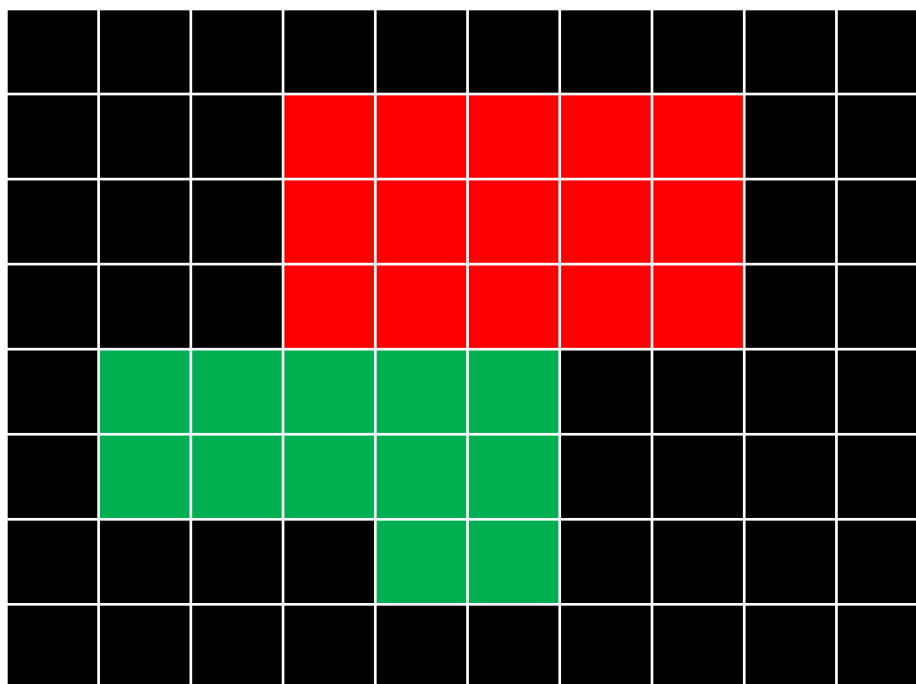
$$\forall p, q \in S, \exists path | p \rightarrow q \subset S$$

Connected Set (Region)



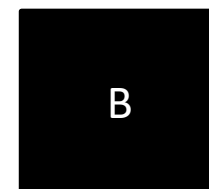
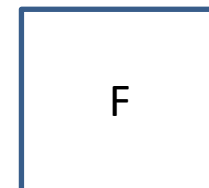
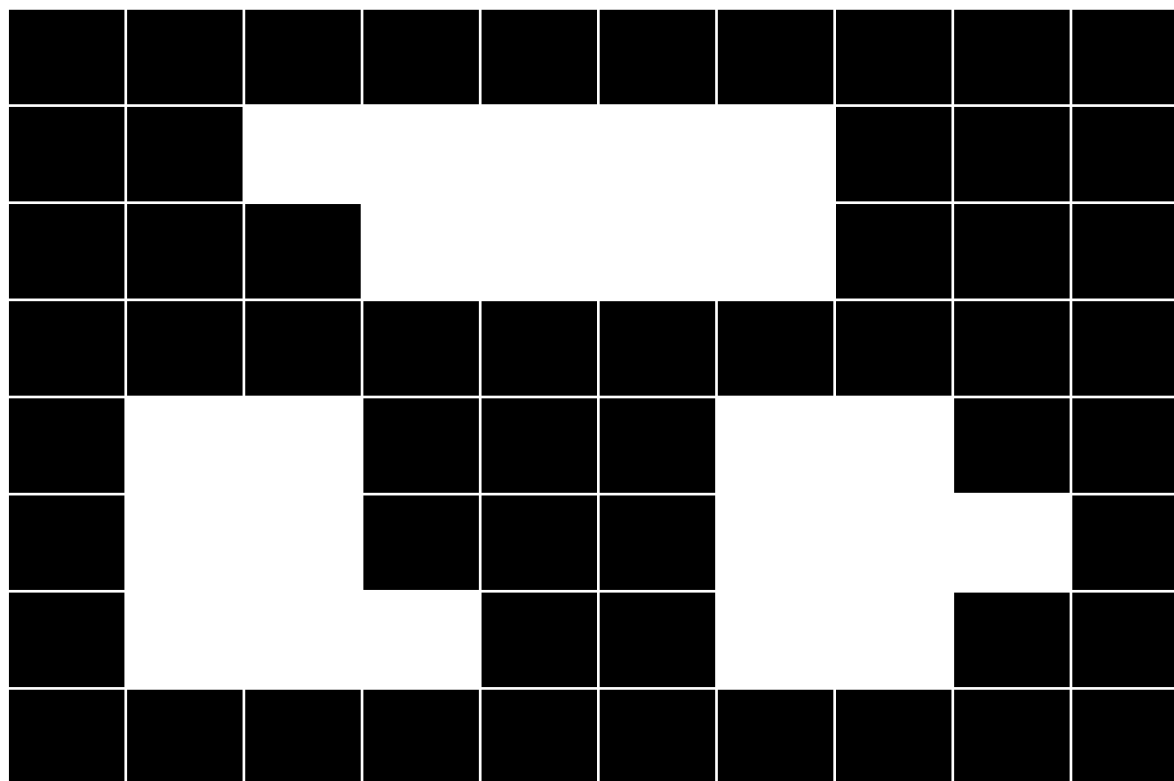
Adjacent Regions

- If $R_i \cup R_j$ is a connected set, R_i and R_j are adjacent regions.



- [illegible]

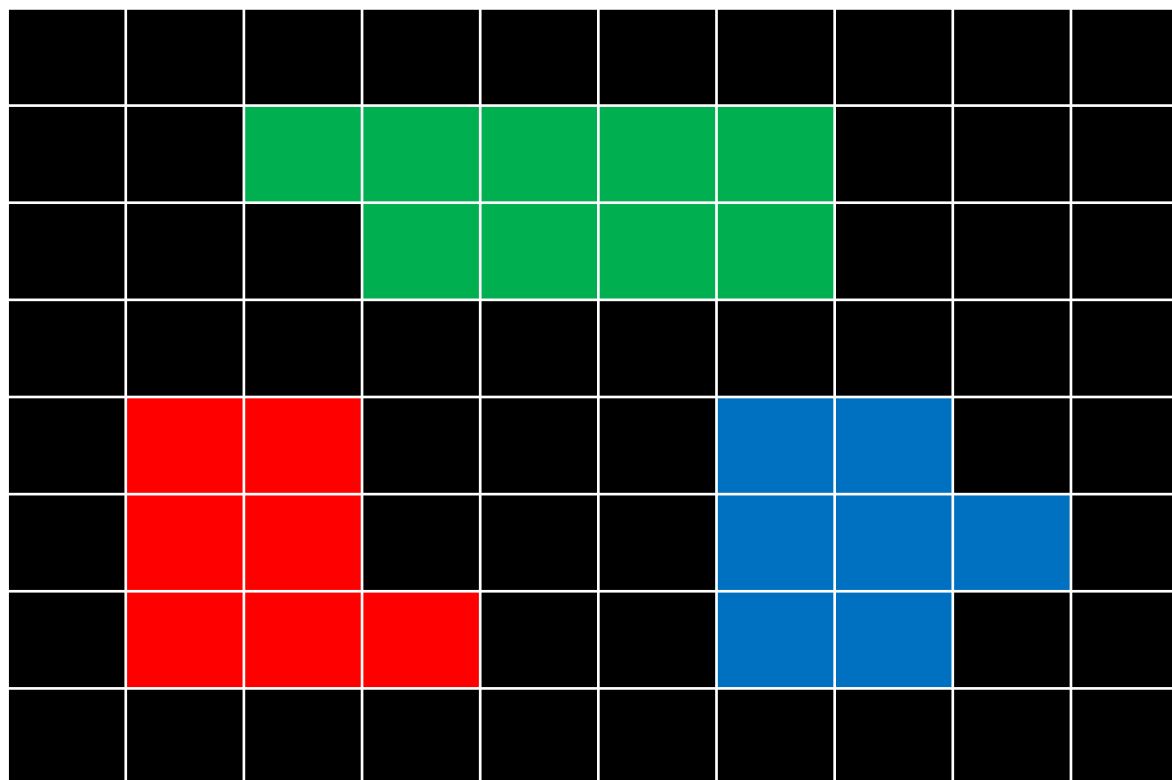
Foreground / Background



Foreground / Background

- K disjoint regions, R_k , $k=1,2,\dots,K$
- $R_u = R_1 \cup R_2 \cup \dots \cup R_K$
- $(R_u)^c$, c means complement
 - Foreground : R_u
 - Background: $(R_u)^c$

Foreground / Background

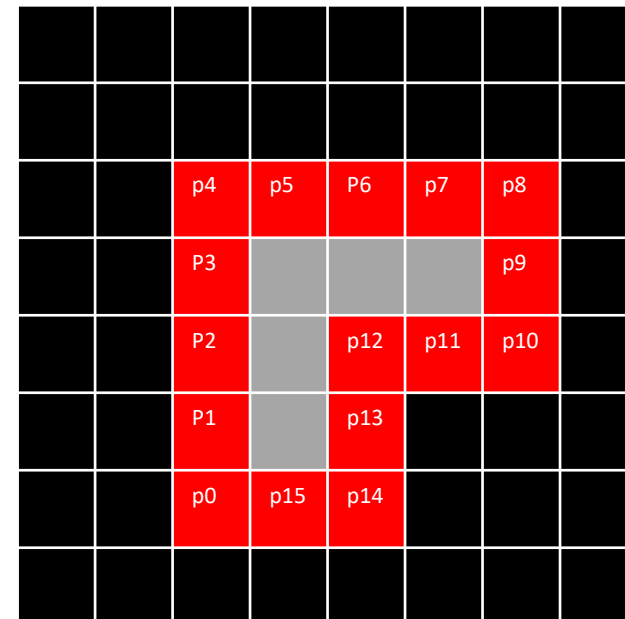
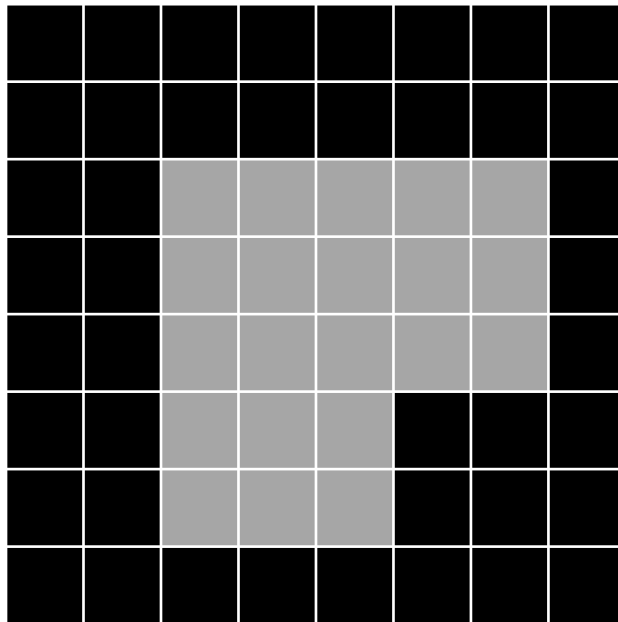


F : Ru

B: (Ru)c

Boundary (Border /Contour) of Region

- Inner Border of Region
 - $B(in) = \{(x,y) \mid (x,y) \in R, (x,y) \text{ adjacent to } (R)^c\}$

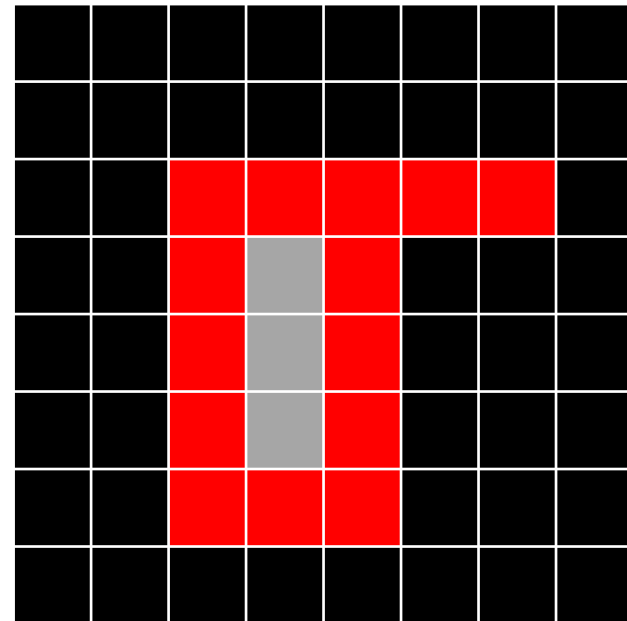


Exercise:

byte[,] getInnerBorder(byte[,]f)

- Input: f is a binary image, 0 means background and 255 means foreground (regions)
- Output: binary image. 255 means inner borders of the regions in input image

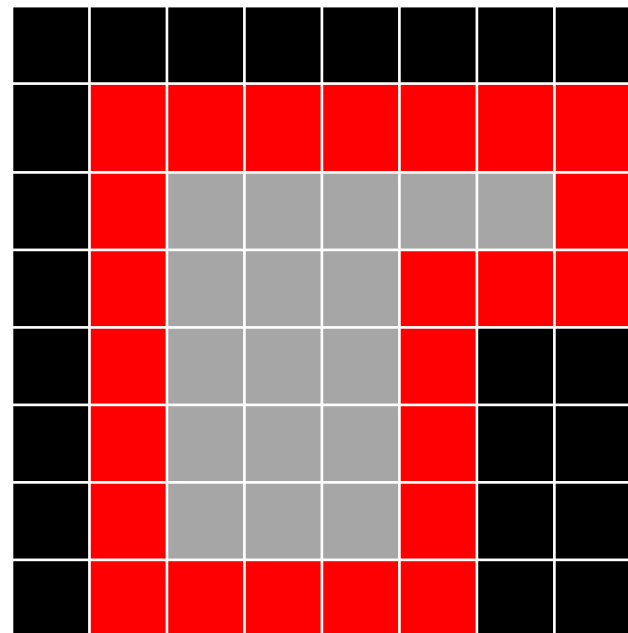
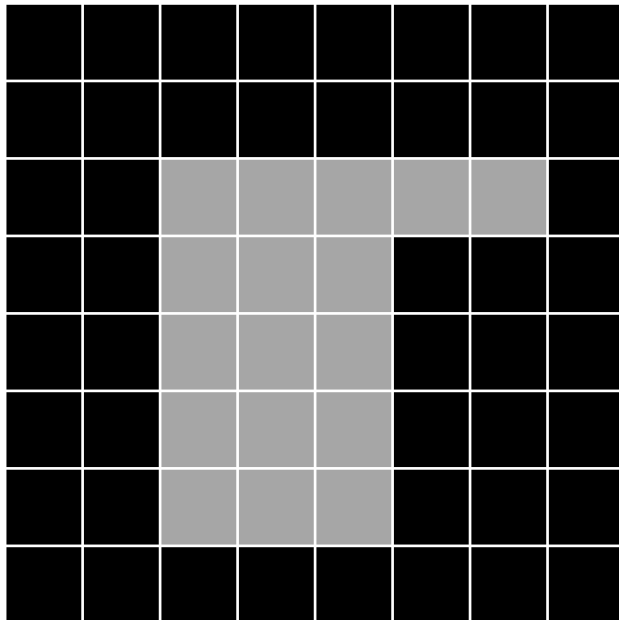
- Inner Border of Region



Exercise:

`byte[,] getOuterBorder(byte[,]f)`

- Outer Border of Region
 - $B(out) = \{(x,y) \mid (x,y) \in (R)^c, (x,y) \text{ adjacent to } R\}$



Exercise:

byte[,] getOuterBorder(byte[,]f)

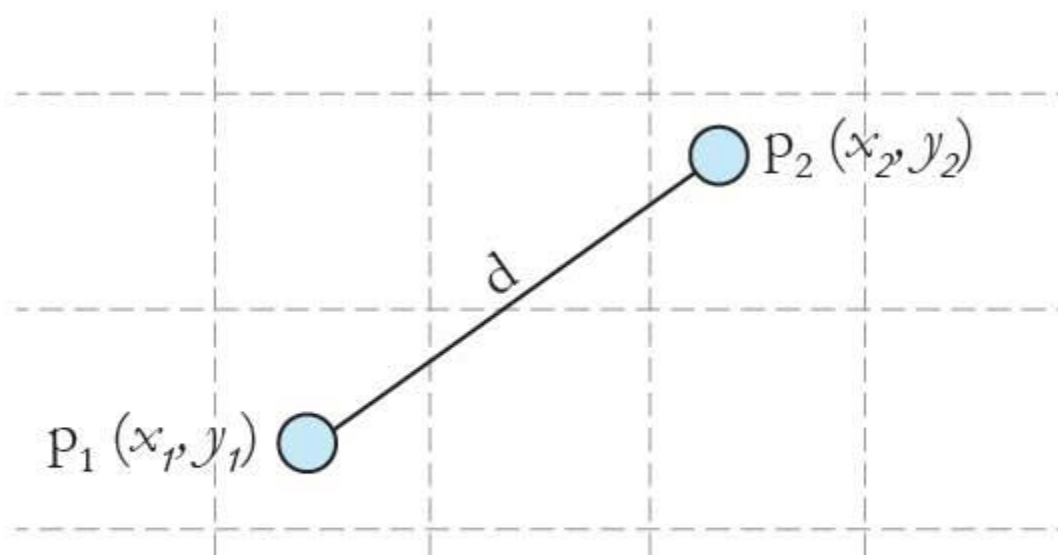
- Input: f is a binary image, 0 means background and 255 means foreground (regions)
- Output: binary image. 255 means outer borders of the regions in input image

2.5.3 Distance Measure

- Distance Function
 - $p(x,y)$, $q(s,t)$, $z(u,v)$
 - (a) $D(p,q) \geq 0$, $D(p,q) = 0$ iff $p = q$
 - (b) $D(p,q) = D(q,p)$
 - (c) $D(p,z) \leq D(p,q) + D(q,z)$

2.5.3 Distance Measure

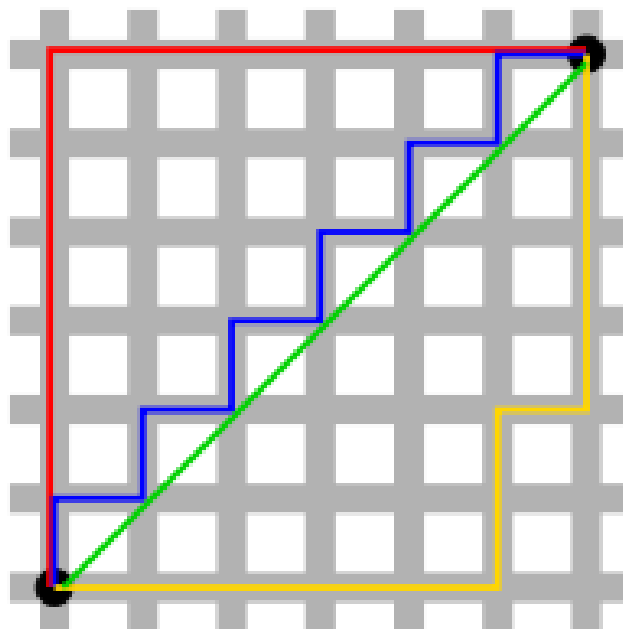
- Euclidean distance [Eu'cl[ai]dean]



$$\text{Euclidean distance } (d) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

2.5.3 Distance Measure

- City-block distance
 - $D_4(p, q) = |x-s| + |y-t|$; $p:(x,y)$, $q(s,t)$

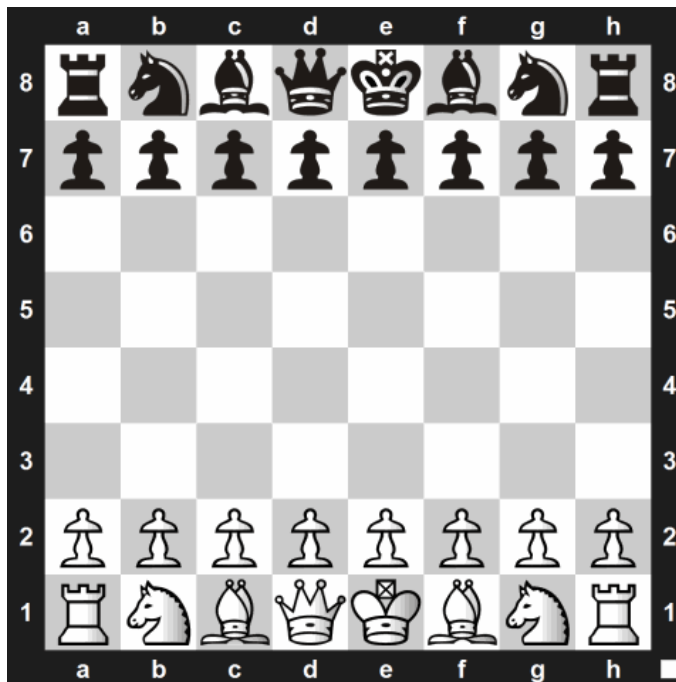


2.5.3 Distance Measure

- Chessboard distance

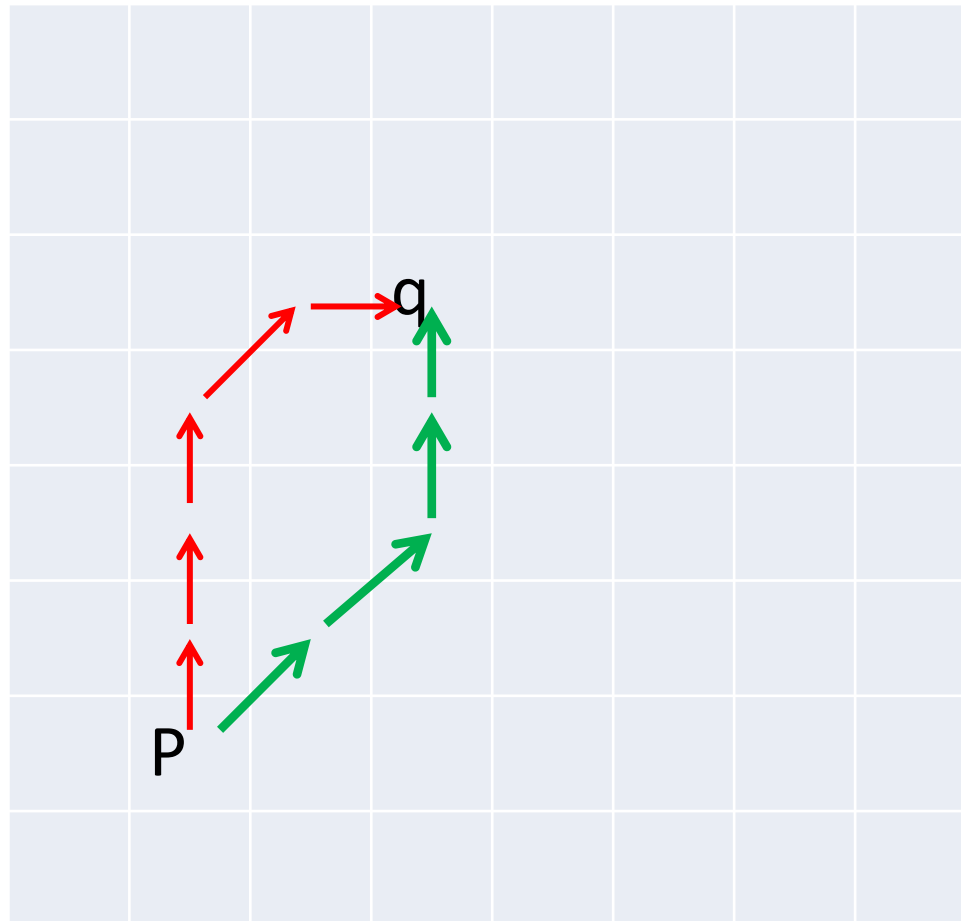
$$D_8(p, q) = \max(|x-s|, |y-t|)$$

chessman: king



2.5.3 Distance Measure

- Chessboard distance



Exercise

- $\text{Max}(D_e, D_4, D_8) = ?$
- $\text{Min}(D_e, D_4, D_8) = ?$

- $D_4(p, q) = |x-s| + |y-t|$
- $D_8(p, q) = \max(|x-s|, |y-t|)$

Exercise

- If $D_4(p, q) \leq K D_8(p, q)$ is always TRUE, what is the min value of K ?
- $D_4(p, q) = |x-s| + |y-t|$
- $D_8(p, q) = \max(|x-s|, |y-t|)$

Exercise

- $p: (x_0, y_0)$
- $R_e = \{ q \mid D_e(p, q) \leq r \}$
- $R_4 = \{ q \mid D_4(p, q) \leq r \}$
- $R_8 = \{ q \mid D_8(p, q) \leq r \}$
 - Draw the sketch of R_e, R_4 & R_8